

VISVESVARAYA TECHNOLOGICAL UNIVERSITY

“JnanaSangama”, Belgaum -590014, Karnataka.



LAB REPORT
on

Machine Learning (23CS6PCMAL)

Submitted by

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in partial fulfillment for the award of the degree of

BACHELOR OF ENGINEERING
in
COMPUTER SCIENCE AND ENGINEERING



B.M.S. COLLEGE OF ENGINEERING

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CERTIFICATE

This is to certify that the Lab work entitled “Machine Learning (23CS6PCMAL)” carried out by **Karan Suresh Nair (1BM23CS139)**, who is bonafide student of **B.M.S. College of Engineering**. It is in partial fulfillment for the award of **Bachelor of Engineering in Computer Science and Engineering** of the Visvesvaraya Technological University, Belgaum. The Lab report has been approved as it satisfies the academic requirements in respect of an Machine Learning (23CS6PCMAL) work prescribed for the said degree.

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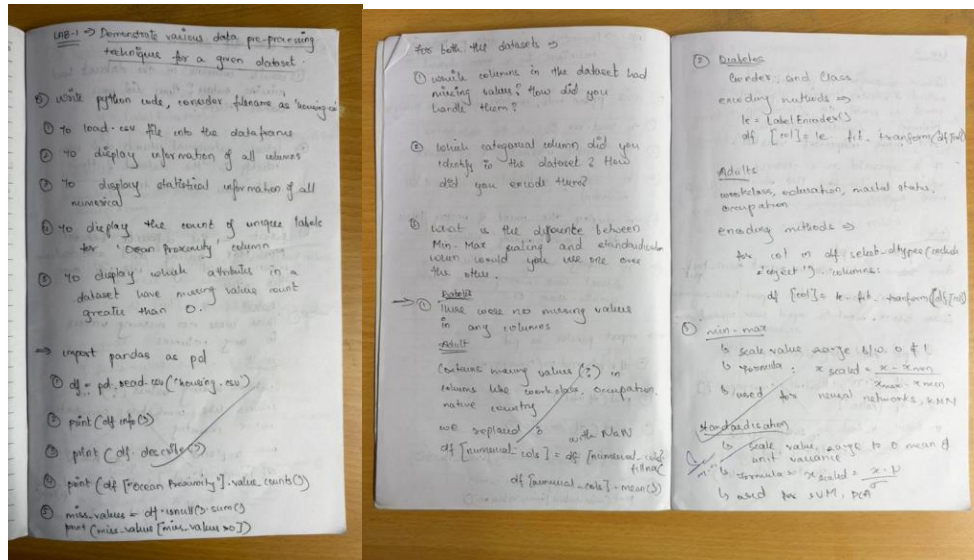
Github Link:

<https://github.com/KaranSureshNair/6thSem-ML-Lab>

Program 1

Write a python program to import and export data using Pandas library functions

Screenshot



Code:

```
import pandas as pd
# Load dataset
housing = pd.read_csv("housing.csv")
# Display first 5 rows
housing.head()
housing.info()
housing.describe()
import matplotlib.pyplot as plt

housing.hist(bins=50, figsize=(15,10))
plt.show()

from sklearn.model_selection import train_test_split

train_set, test_set = train_test_split(housing, test_size=0.2, random_state=42)

import numpy as np

housing["income_cat"] = pd.cut(housing["median_income"],
                               bins=[0., 1.5, 3.0, 4.5, 6., np.inf],
```

```

labels=[1,2,3,4,5])

from sklearn.model_selection import StratifiedShuffleSplit

split = StratifiedShuffleSplit(n_splits=1, test_size=0.2, random_state=42)

for train_index, test_index in split.split(housing, housing["income_cat"]):
    strat_train_set = housing.loc[train_index]
    strat_test_set = housing.loc[test_index]

housing.plot(kind="scatter", x="longitude", y="latitude",
              alpha=0.4,
              s=housing["population"]/100,
              label="population",
              c="median_house_value",
              cmap="jet",
              colorbar=True)
plt.legend()
plt.show()

corr_matrix = housing.corr(numeric_only=True)

corr_matrix["median_house_value"].sort_values(ascending=False)

housing.plot(kind="scatter", x="median_income", y="median_house_value", alpha=0.1)
plt.show()

housing["rooms_per_household"] = housing["total_rooms"] / housing["households"]
housing["bedrooms_per_room"] = housing["total_bedrooms"] / housing["total_rooms"]
housing["population_per_household"] = housing["population"] / housing["households"]

corr_matrix = housing.corr(numeric_only=True)
corr_matrix["median_house_value"].sort_values(ascending=False)

housing["total_bedrooms"].isnull().sum()
from sklearn.impute import SimpleImputer

imputer = SimpleImputer(strategy="median")

housing_num = housing.drop("ocean_proximity", axis=1)

imputer.fit(housing_num)

housing_num_imputed = imputer.transform(housing_num)

housing["ocean_proximity"].value_counts()
from sklearn.preprocessing import OneHotEncoder

```

```

housing_cat = housing[["ocean_proximity"]]

encoder = OneHotEncoder()

housing_cat_encoded = encoder.fit_transform(housing_cat)

housing_cat_encoded.toarray()

from sklearn.preprocessing import StandardScaler

scaler = StandardScaler()

housing_scaled = scaler.fit_transform(housing_num_imputed)

from sklearn.pipeline import Pipeline
from sklearn.compose import ColumnTransformer
from sklearn.base import BaseEstimator, TransformerMixin

# Custom Transformer
class CombinedAttributesAdder(BaseEstimator, TransformerMixin):
    def __init__(self, add_bedrooms_per_room=True):
        self.add_bedrooms_per_room = add_bedrooms_per_room

    def fit(self, X, y=None):
        return self

    def transform(self, X):
        rooms_per_household = X[:, 3] / X[:, 6]
        population_per_household = X[:, 5] / X[:, 6]

        if self.add_bedrooms_per_room:
            bedrooms_per_room = X[:, 4] / X[:, 3]
            return np.c_[X, rooms_per_household, population_per_household, bedrooms_per_room]
        else:
            return np.c_[X, rooms_per_household, population_per_household]

num_attribs = list(housing_num.columns)
cat_attribs = ["ocean_proximity"]

num_pipeline = Pipeline([
    ('imputer', SimpleImputer(strategy="median")),
    ('attribs_adder', CombinedAttributesAdder()),
    ('std_scaler', StandardScaler())
])

full_pipeline = ColumnTransformer([
    ("num", num_pipeline, num_attribs),

```

```
    ("cat", OneHotEncoder(), cat_attribs)
])

housing_prepared = full_pipeline.fit_transform(housing)

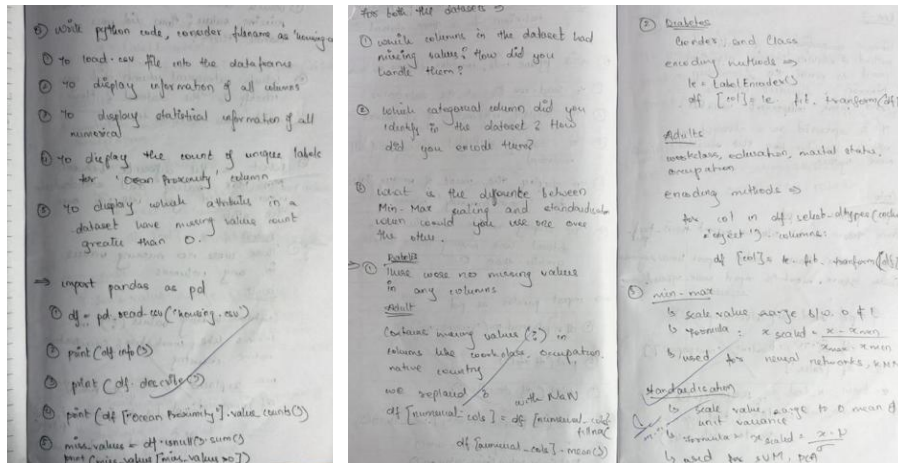
from sklearn.linear_model import LinearRegression

lin_reg = LinearRegression()
lin_reg.fit(housing_prepared, housing["median_house_value"])
```

Program 2

Demonstrate various data pre-processing techniques for a given dataset

Screenshot



Code:

```
import pandas as pd
import numpy as np
import matplotlib.pyplot as plt
from sklearn.model_selection import train_test_split
from sklearn.impute import SimpleImputer
from sklearn.preprocessing import OrdinalEncoder, OneHotEncoder
from sklearn.preprocessing import StandardScaler, MinMaxScaler
from scipy import stats

def createdata():
    data = {
        'Age': np.random.randint(18, 70, size=20),
        'Salary': np.random.randint(30000, 120000, size=20),
        'Purchased': np.random.choice([0, 1], size=20),
        'Gender': np.random.choice(['Male', 'Female'], size=20),
        'City': np.random.choice(['New York', 'San Francisco', 'Los Angeles'], size=20)
    }

    df = pd.DataFrame(data)
    return df
df = createdata()
df.head(10)
```



```

df.shape
# Introduce some missing values for demonstration
df.loc[5, 'Age'] = np.nan
df.loc[10, 'Salary'] = np.nan
df.head(10)
# Basic information about the dataset
print(df.info())
# Summary statistics
print(df.describe())
# Code to Find Missing Values
# Check for missing values in each column
missing_values = df.isnull().sum()

# Display columns with missing values
print(missing_values[missing_values > 0])
# Set the values to some value (zero, the mean, the median, etc.).
# Step 1: Create an instance of SimpleImputer with the median strategy for Age and mean strategy
for Salary
imputer1 = SimpleImputer(strategy="median")
imputer2 = SimpleImputer(strategy="mean")

df_copy=df

# Step 2: Fit the imputer on the "Age" and "Salary" column
# Note: SimpleImputer expects a 2D array, so we reshape the column
imputer1.fit(df_copy[["Age"]])
imputer2.fit(df_copy[["Salary"]])

# Step 3: Transform (fill) the missing values in the "Age" and "Salary" column
df_copy["Age"] = imputer1.transform(df[["Age"]])
df_copy["Salary"] = imputer2.transform(df[["Salary"]])

# Verify that there are no missing values left
print(df_copy["Age"].isnull().sum())
print(df_copy["Salary"].isnull().sum())
# Handling Categorical Attributes
# Using Ordinal Encoding for gender Column and One-Hot Encoding for City Column

# Initialize OrdinalEncoder
ordinal_encoder = OrdinalEncoder(categories=[["Male", "Female"]])
# Fit and transform the data
df_copy["Gender_Encoded"] = ordinal_encoder.fit_transform(df_copy[["Gender"]])

# Initialize OneHotEncoder
onehot_encoder = OneHotEncoder()

# Fit and transform the "City" column
encoded_data = onehot_encoder.fit_transform(df[["City"]])

```

```

# Convert the sparse matrix to a dense array
encoded_array = encoded_data.toarray()

# Convert to DataFrame for better visualization
encoded_df = pd.DataFrame(encoded_array,
columns=onehot_encoder.get_feature_names_out(["City"]))
df_encoded = pd.concat([df_copy, encoded_df], axis=1)

df_encoded.drop("Gender", axis=1, inplace=True)
df_encoded.drop("City", axis=1, inplace=True)

print(df_encoded.head())
#Data Transformation
# Min-Max Scaler/Normalization (range 0-1)
#Pros: Keeps all data between 0 and 1; ideal for distance-based models.
#Cons: Can distort data distribution, especially with extreme outliers.
normalizer = MinMaxScaler()
df_encoded[['Salary']] = normalizer.fit_transform(df_encoded[['Salary']])
df_encoded.head()
# Standardization (mean=0, variance=1)
#Pros: Works well for normally distributed data; suitable for many models.
#Cons: Sensitive to outliers.
scaler = StandardScaler()
df_encoded[['Age']] = scaler.fit_transform(df_encoded[['Age']])
df_encoded.head()

#Removing Outliers
# Outlier Detection and Treatment using IQR
#Pros: Simple and effective for mild outliers.
#Cons: May overly reduce variation if there are many extreme outliers.
df_encoded_copy1=df_encoded
df_encoded_copy2=df_encoded
df_encoded_copy3=df_encoded

Q1 = df_encoded_copy1['Salary'].quantile(0.25)
Q3 = df_encoded_copy1['Salary'].quantile(0.75)
IQR = Q3 - Q1
lower_bound = Q1 - 1.5 * IQR
upper_bound = Q3 + 1.5 * IQR
df_encoded_copy1['Salary'] = np.where(df_encoded_copy1['Salary'] > upper_bound, upper_bound,
np.where(df_encoded_copy1['Salary'] < lower_bound, lower_bound,
df_encoded_copy1['Salary']))

print(df_encoded_copy1.head())

#Removing Outliers

```

Z-score method

#Pros: Good for normally distributed data.

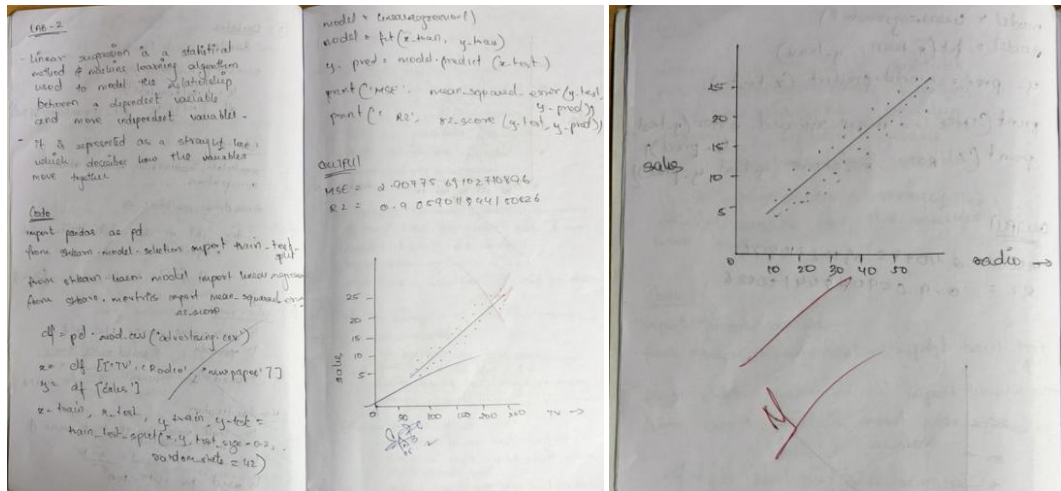
#Cons: Not suitable for non-normal data; may miss outliers in skewed distributions.

```
df_encoded_copy2['Salary_zscore'] = stats.zscore(df_encoded_copy2['Salary'])
df_encoded_copy2['Salary'] = np.where(df_encoded_copy2['Salary_zscore'].abs() > 3, np.nan,
df_encoded_copy2['Salary']) # Replace outliers with NaN
print(df_encoded_copy2.head())
#Removing Outliers
# Median replacement for outliers
#Pros: Keeps distribution shape intact, useful when capping isn't feasible.
#Cons: May distort data if outliers represent real phenomena.
df_encoded_copy3['Salary_zscore'] = stats.zscore(df_encoded_copy3['Salary'])
median_salary = df_encoded_copy3['Salary'].median()
df_encoded_copy3['Salary'] = np.where(df_encoded_copy3['Salary_zscore'].abs() > 3,
median_salary, df_encoded_copy3['Salary'])
print(df_encoded_copy3.head())
```

Program 3

Implement Linear and Multi-Linear Regression algorithm using appropriate dataset

Screenshot



Code:

```
# Predict Canada's Per Capita Income for 2020
```

```
# Import libraries
```

```
import pandas as pd
```

```
import numpy as np
```

```
import matplotlib.pyplot as plt
```

```
from sklearn.linear_model import LinearRegression
```

```
# Load dataset
```

```
df = pd.read_csv('canada_per_capita_income.csv')
```

```
# Display first 5 rows
```

```
print(df.head())
```

```
# Define independent and dependent variables
```

```
X = df[['year']] # Feature
```

```
y = df['per capita income (US$)']
```

```
# Create model
```

```
model = LinearRegression()
```

```
# Train model
```

```

model.fit(X, y)

# Predict income for year 2020
predicted_income = model.predict([[2020]])

print("Predicted Per Capita Income in 2020:", predicted_income[0])

plt.scatter(X, y)
plt.plot(X, model.predict(X))
plt.xlabel("Year")
plt.ylabel("Per Capita Income")
plt.show()
# -----
# Import libraries
import pandas as pd
import matplotlib.pyplot as plt
from sklearn.linear_model import LinearRegression
# -----
# Load dataset
df = pd.read_csv('salary.csv')

# Remove missing values
df.dropna(inplace=True)

# Define variables
X = df[['YearsExperience']]
y = df['Salary']

# Create and train model
model = LinearRegression()
model.fit(X, y)

# Predict salary for 12 years experience
predicted_salary = model.predict([[12]])

print("Predicted Salary for 12 years experience:", predicted_salary[0])

# ♦ Plot graph
plt.scatter(X, y)          # Actual data points
plt.plot(X, model.predict(X)) # Regression line

plt.xlabel("Years of Experience")
plt.ylabel("Salary")
plt.title("Salary vs Experience (Linear Regression)")

plt.show()
# Import libraries

```

```

import pandas as pd
from sklearn.linear_model import LinearRegression
# -----
# Load dataset
df = pd.read_csv('hiring.csv')

# Convert word numbers to numeric
df['experience'] = df['experience'].replace({
    'zero':0,'one':1,'two':2,'three':3,'four':4,
    'five':5,'six':6,'seven':7,'eight':8,'nine':9,'ten':10
})

# Convert to numeric type
df['experience'] = pd.to_numeric(df['experience'], errors='coerce')

# Fill missing values using median
df['experience'] = df['experience'].fillna(df['experience'].median())
df['test_score(out of 10)'] = df['test_score(out of 10)'].fillna(
    df['test_score(out of 10)'].median()
)

X = df[['experience',
        'test_score(out of 10)',
        'interview_score(out of 10)']]

y = df['salary($)']

model = LinearRegression()
model.fit(X, y)

# Candidate 1
input1 = pd.DataFrame([[2, 9, 6]],
                       columns=X.columns)
salary1 = model.predict(input1)

# Candidate 2
input2 = pd.DataFrame([[12, 10, 10]],
                       columns=X.columns)
salary2 = model.predict(input2)

print("Predicted Salary (2 yr exp, 9 test, 6 interview):", salary1[0])
print("Predicted Salary (12 yr exp, 10 test, 10 interview):", salary2[0])
# 1000 Companies Profit Prediction
# Import libraries
import pandas as pd
from sklearn.linear_model import LinearRegression
# -----
# Load dataset

```

```

df = pd.read_csv('1000_Companies.csv')

# Handle categorical column (State)
df = pd.get_dummies(df, columns=['State'], drop_first=True)

# Define features and target
X = df.drop('Profit', axis=1)
y = df['Profit']

model = LinearRegression()
model.fit(X, y)

# Create input for Florida
# (Assuming columns are: R&D Spend, Administration, Marketing Spend,
# State_Florida, State_New York)

input_profit = pd.DataFrame([[91694.48,
                             515841.3,
                             11931.24,
                             1, 0]], # Florida = 1
                             columns=X.columns)

predicted_profit = model.predict(input_profit)

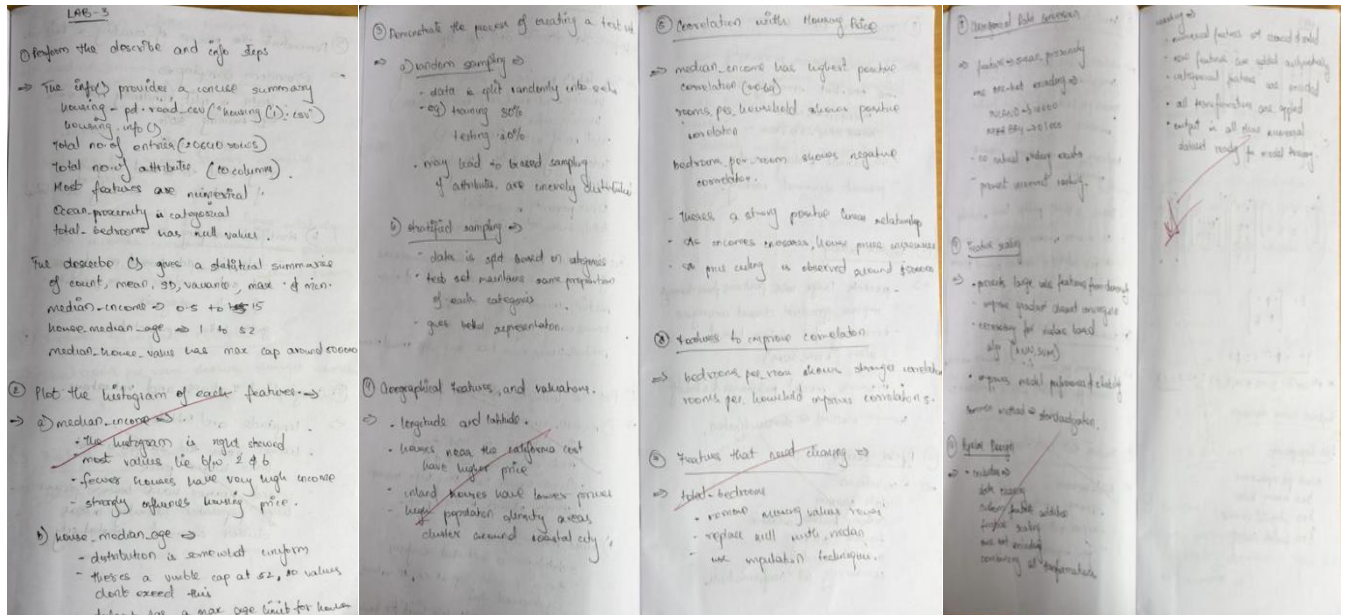
print("Predicted Profit:", predicted_profit[0])

```

Program 4

Build Logistic Regression Model for a given dataset

Screenshot



Code:

Logistic regression - binary classification

Import Libraries

import pandas as pd

import numpy as np

import matplotlib.pyplot as plt

from sklearn.model_selection import train_test_split

from sklearn.linear_model import LogisticRegression

from sklearn.metrics import accuracy_score, classification_report

Load Dataset

df = pd.read_csv("HR_comma_sep.csv")

1. Exploratory Data Analysis

print("First 5 rows:")

print(df.head())

print("\nDataset Info:")


```

print(df.info())

print("\nEmployee Retention Count:")
print(df['left'].value_counts())

print("\nAverage values grouped by retention:")
print(df.groupby('left').mean(numeric_only=True))

# -----
# 2. Salary vs Retention (Bar Chart)
# -----

pd.crosstab(df.salary, df.left).plot(kind='bar')
plt.title("Salary vs Employee Retention")
plt.xlabel("Salary Level")
plt.ylabel("Number of Employees")
plt.show()

# -----
# 3. Department vs Retention (Bar Chart)
# -----

pd.crosstab(df.Department, df.left).plot(kind='bar', figsize=(10,5))
plt.title("Department vs Employee Retention")
plt.xlabel("Department")
plt.ylabel("Number of Employees")
plt.xticks(rotation=45)
plt.show()

# -----
# 4. Feature Selection
# -----
# Selecting important features
df2 = df[['satisfaction_level',
          'average_monthly_hours',
          'promotion_last_5years',
          'salary',
          'left']]

# Convert salary (categorical) to dummy variables
df2 = pd.get_dummies(df2, columns=['salary'], drop_first=True)

# -----
# 5. Train-Test Split
# -----

X = df2.drop('left', axis=1)
y = df2['left']

X_train, X_test, y_train, y_test = train_test_split(

```

```

X, y, test_size=0.3, random_state=42)

# -----
# 6. Build Logistic Regression Model
# -----

model = LogisticRegression(max_iter=1000)
model.fit(X_train, y_train)

# -----
# 7. Prediction & Accuracy
# -----

y_pred = model.predict(X_test)

accuracy = accuracy_score(y_test, y_pred)

print("\nModel Accuracy:", accuracy)
print("\nClassification Report:\n", classification_report(y_test, y_pred))

# Logistic regression - multiclass classification
# Logistic Regression - Multiclass Classification (Zoo Dataset)

# Import Libraries
import pandas as pd
import numpy as np
import matplotlib.pyplot as plt
from sklearn.model_selection import train_test_split
from sklearn.linear_model import LogisticRegression
from sklearn.metrics import accuracy_score, confusion_matrix, ConfusionMatrixDisplay

# -----
# 1. Load Dataset
# -----

zoo = pd.read_csv("zoo-data.csv")
class_type = pd.read_csv("zoo-class-type.csv")

print("First 5 rows:")
print(zoo.head())

# -----
# 2. Data Preprocessing
# -----

# Drop animal name (not useful for prediction)
zoo = zoo.drop("animal_name", axis=1)

```

```

# Check for null values
print("\nMissing values:")
print(zoo.isnull().sum())

# Features and Target
X = zoo.drop("class_type", axis=1)
y = zoo["class_type"]

# -----
# 3. Train-Test Split
# -----

X_train, X_test, y_train, y_test = train_test_split(
    X, y, test_size=0.3, random_state=42)

# -----
# 4. Build Logistic Regression Model
# -----

model = LogisticRegression(max_iter=1000, multi_class='multinomial')
model.fit(X_train, y_train)

# -----
# 5. Prediction
# -----

y_pred = model.predict(X_test)

# -----
# 6. Accuracy
# -----

accuracy = accuracy_score(y_test, y_pred)
print("\nModel Accuracy:", accuracy)

# -----
# 7. Confusion Matrix
# -----

cm = confusion_matrix(y_test, y_pred)

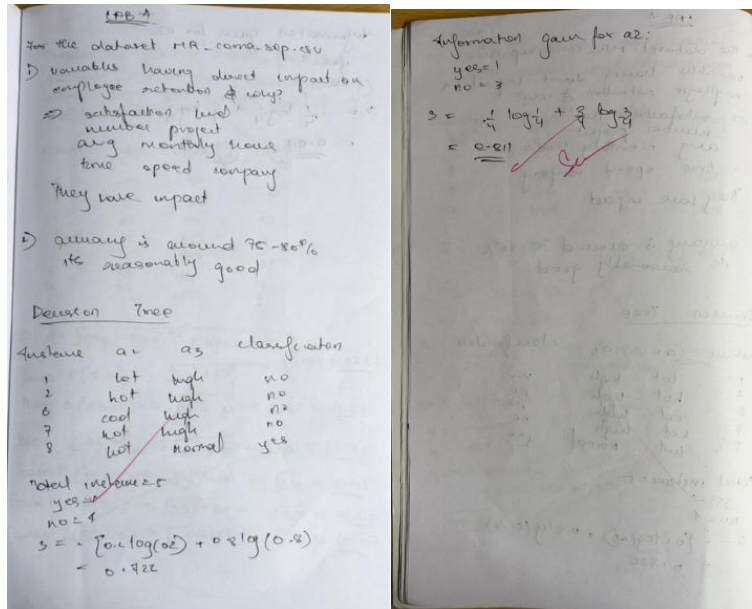
disp = ConfusionMatrixDisplay(confusion_matrix=cm)
disp.plot()
plt.title("Confusion Matrix - Zoo Dataset")
plt.show()

```

Program 5

Use an appropriate data set for building the decision tree (ID3) and apply this knowledge to classify a new sample.

Screenshot



Code:

Decision Tree - IRIS Dataset

Import Libraries

import pandas as pd

import numpy as np

from sklearn.model_selection import train_test_split

from sklearn.tree import DecisionTreeClassifier

from sklearn.metrics import accuracy_score, confusion_matrix, ConfusionMatrixDisplay

import matplotlib.pyplot as plt

Load Dataset

iris = pd.read_csv("iris.csv")

print("First 5 rows:")

print(iris.head())

```

# -----
# Features and Target
# -----
X = iris.drop("species", axis=1) # change column name if different
y = iris["species"]

# -----
# Train-Test Split (80-20)
# -----
X_train, X_test, y_train, y_test = train_test_split(
    X, y, test_size=0.2, random_state=42)

# -----
# Build Model
# -----
model = DecisionTreeClassifier()
model.fit(X_train, y_train)

# -----
# Prediction
# -----
y_pred = model.predict(X_test)

# -----
# Accuracy
# -----
accuracy = accuracy_score(y_test, y_pred)
print("\nIRIS Dataset Accuracy:", accuracy)

# -----
# Confusion Matrix
# -----
cm = confusion_matrix(y_test, y_pred)
disp = ConfusionMatrixDisplay(confusion_matrix=cm)
disp.plot()
plt.title("Confusion Matrix - IRIS Dataset")
plt.show()

```

Decision Tree - Drug Dataset

```

# Import Libraries
import pandas as pd
import numpy as np
from sklearn.model_selection import train_test_split
from sklearn.tree import DecisionTreeClassifier
from sklearn.preprocessing import LabelEncoder

```

```

from sklearn.metrics import accuracy_score, confusion_matrix, ConfusionMatrixDisplay
import matplotlib.pyplot as plt

# -----
# Load Dataset
# -----
drug = pd.read_csv("drug.csv")

print("First 5 rows:")
print(drug.head())

# -----
# Data Preprocessing
# -----

# Encode categorical columns
le = LabelEncoder()

for column in drug.columns:
    if drug[column].dtype == 'object':
        drug[column] = le.fit_transform(drug[column])

# -----
# Features and Target
# -----
X = drug.drop("Drug", axis=1) # change if column name different
y = drug["Drug"]

# -----
# Train-Test Split (80-20)
# -----
X_train, X_test, y_train, y_test = train_test_split(
    X, y, test_size=0.2, random_state=42)

# -----
# Build Model
# -----
model = DecisionTreeClassifier()
model.fit(X_train, y_train)

# -----
# Prediction
# -----
y_pred = model.predict(X_test)

# -----
# Accuracy
# -----

```

```

accuracy = accuracy_score(y_test, y_pred)
print("\nDrug Dataset Accuracy:", accuracy)

# -----
# Confusion Matrix
# -----
cm = confusion_matrix(y_test, y_pred)
disp = ConfusionMatrixDisplay(confusion_matrix=cm)
disp.plot()
plt.title("Confusion Matrix - Drug Dataset")
plt.show()

```

Regression Tree - Petrol Consumption Dataset

```

# Import Libraries
import pandas as pd
import numpy as np
from sklearn.model_selection import train_test_split
from sklearn.tree import DecisionTreeRegressor
from sklearn.metrics import mean_absolute_error, mean_squared_error

# -----
# Load Dataset
# -----
data = pd.read_csv("petrol_consumption.csv")

print("First 5 rows:")
print(data.head())

# -----
# Features and Target
# -----
# Assuming last column is Petrol_Consumption
X = data.iloc[:, :-1] # All columns except last
y = data.iloc[:, -1] # Last column (target)

# -----
# Train-Test Split (80% - 20%)
# -----
X_train, X_test, y_train, y_test = train_test_split(
    X, y, test_size=0.2, random_state=42)

# -----
# Build Regression Tree Model
# -----
regressor = DecisionTreeRegressor(random_state=42)
regressor.fit(X_train, y_train)

```

```

# -----
# Prediction
# -----
y_pred = regressor.predict(X_test)

# -----
# Evaluation Metrics
# -----
mae = mean_absolute_error(y_test, y_pred)
mse = mean_squared_error(y_test, y_pred)
rmse = np.sqrt(mse)

print("\nModel Evaluation Metrics:")
print("Mean Absolute Error (MAE):", mae)
print("Mean Squared Error (MSE):", mse)
print("Root Mean Squared Error (RMSE):", rmse)

```


Program 6

Build KNN Classification model for a given dataset.

Screenshot

LAB-6

Q) Consider the following dataset, for $k=3$ of test data (v, ss, ice) as (Person, Age, Salary) solve using KNN classifier model & predict the target.

Person	Age	Salary	Target
A	18	50	N
B	23	55	N
C	24	40	N
D	41	60	Y
E	45	40	Y
F	38	40	Y
X	35	100	?

$d = \sqrt{(Age - 10)^2 + (Salary - 50)^2}$

Person A (18, 50) $\rightarrow \sqrt{17^2 + 50^2} = \sqrt{2789} \approx 52.82$

Person B (23, 55) $\rightarrow \sqrt{12^2 + 45^2} = \sqrt{169} \approx 46.57$

Person C (24, 40) $\rightarrow \sqrt{11^2 + 30^2} = \sqrt{1021} \approx 31.95$

Person D (41, 60) $\rightarrow \sqrt{31^2 + 10^2} = \sqrt{1021} \approx 31.95$

Person E (45, 40) $\rightarrow \sqrt{35^2 + 30^2} = \sqrt{1669} \approx 40.85$

Person F (38, 40) $\rightarrow \sqrt{28^2 + 30^2} = \sqrt{1604} \approx 40.07$

Step 2: Sort by Distance $\rightarrow$

Person	Distance	Target
E	31.95	Y
C	31.95	N
D	40.85	Y
B	46.57	N
A	52.82	N
F	60.07	Y

Step 3: select $k=3$ nearest Neighbors

Person	Target
E	Y
C	N
D	Y

Vote: Y=2, N=1

Majority class = Y

So predicted target for $k=3$

For the dataset - How to choose k value?

$\rightarrow$ The value of K in KNN determines how many nearest neighbors are considered when classifying a data point.

To choose the optimal K value:

- 1) Train the KNN model with $k(1-20)$
- 2) Compute accuracy rate & error rate
- 3) Plot accuracy vs K & Error vs K.
- 4) Choose the K value where:
 - Accuracy is maximum
 - Error rate is minimum.

$\Rightarrow$ best K is the point where the curve stable

For Distance Based - Purpose of Feature Scaling

$\rightarrow$ Feature scaling is done using standardization.

$$z = \frac{x - p}{\sigma}$$

X $\rightarrow$ feature value
p $\rightarrow$ mean
 σ $\rightarrow$ standard deviation.

It is used to normalize the range of independent variables. Ensures all features contribute equally.

Code:

```
# KNN Classifier for IRIS Dataset
```

```
# Import libraries
```

```
import pandas as pd
```

```
from sklearn.model_selection import train_test_split
```

```
from sklearn.neighbors import KNeighborsClassifier
```

```
from sklearn.metrics import accuracy_score, confusion_matrix, classification_report
```

```
import seaborn as sns
```

```
import matplotlib.pyplot as plt
```

```
# Load dataset
```

```
iris = pd.read_csv("iris.csv")
```

```
# Features and target
```

```
X = iris.drop("species", axis=1) # change if target column name differs
```

```
y = iris["species"]
```

```
# Train-Test Split (80-20)
```

```
X_train, X_test, y_train, y_test = train_test_split(X, y, test_size=0.2, random_state=42)
```

```
# KNN Model
```

```
k = 5
```

```

model = KNeighborsClassifier(n_neighbors=k)
model.fit(X_train, y_train)

# Prediction
y_pred = model.predict(X_test)

# Accuracy
print("Accuracy:", accuracy_score(y_test, y_pred))

# Confusion Matrix
cm = confusion_matrix(y_test, y_pred)
print("Confusion Matrix:\n", cm)

# Classification Report
print("Classification Report:\n", classification_report(y_test, y_pred))

# Plot Confusion Matrix
sns.heatmap(cm, annot=True, fmt='d')
plt.title("Confusion Matrix - Iris Dataset")
plt.show()

# KNN Classifier for Diabetes Dataset (With Feature Scaling)
# Import libraries
import pandas as pd
from sklearn.model_selection import train_test_split
from sklearn.preprocessing import StandardScaler
from sklearn.neighbors import KNeighborsClassifier
from sklearn.metrics import accuracy_score, confusion_matrix

# Load dataset
diabetes = pd.read_csv("diabetes.csv")

# Features and target
X = diabetes.drop("Outcome", axis=1)
y = diabetes["Outcome"]

# Train-Test Split
X_train, X_test, y_train, y_test = train_test_split(X, y, test_size=0.2, random_state=42)

# Feature Scaling
scaler = StandardScaler()
X_train = scaler.fit_transform(X_train)
X_test = scaler.transform(X_test)

# KNN Model
k = 7
model = KNeighborsClassifier(n_neighbors=k)

```

```

model.fit(X_train, y_train)

# Prediction
y_pred = model.predict(X_test)

# Accuracy
print("Accuracy:", accuracy_score(y_test, y_pred))

# Confusion Matrix
cm = confusion_matrix(y_test, y_pred)
print("Confusion Matrix:\n", cm)

# KNN Classifier for Heart Dataset (Finding Best K)
# Import libraries
import pandas as pd
import matplotlib.pyplot as plt
import seaborn as sns
from sklearn.model_selection import train_test_split
from sklearn.preprocessing import StandardScaler
from sklearn.neighbors import KNeighborsClassifier
from sklearn.metrics import accuracy_score, confusion_matrix, classification_report

# Load dataset
heart = pd.read_csv("heart.csv")

# Features and target
X = heart.drop("target", axis=1)
y = heart["target"]

# Train-Test Split
X_train, X_test, y_train, y_test = train_test_split(X, y, test_size=0.2, random_state=42)

# Feature Scaling
scaler = StandardScaler()
X_train = scaler.fit_transform(X_train)
X_test = scaler.transform(X_test)

# Find best K
scores = []

for k in range(1,21):
    model = KNeighborsClassifier(n_neighbors=k)
    model.fit(X_train, y_train)
    pred = model.predict(X_test)
    scores.append(accuracy_score(y_test, pred))

# Plot K vs Accuracy

```

```

plt.plot(range(1,21), scores)
plt.xlabel("K value")
plt.ylabel("Accuracy")
plt.title("Finding Best K")
plt.show()

# Best K
best_k = scores.index(max(scores)) + 1
print("Best K:", best_k)

# Train final model
model = KNeighborsClassifier(n_neighbors=best_k)
model.fit(X_train, y_train)

# Prediction
y_pred = model.predict(X_test)

# Accuracy
print("Accuracy:", accuracy_score(y_test, y_pred))

# Confusion Matrix
cm = confusion_matrix(y_test, y_pred)
print("Confusion Matrix:\n", cm)

# Plot Confusion Matrix
sns.heatmap(cm, annot=True, fmt='d')
plt.title("Confusion Matrix - Heart Dataset")
plt.show()

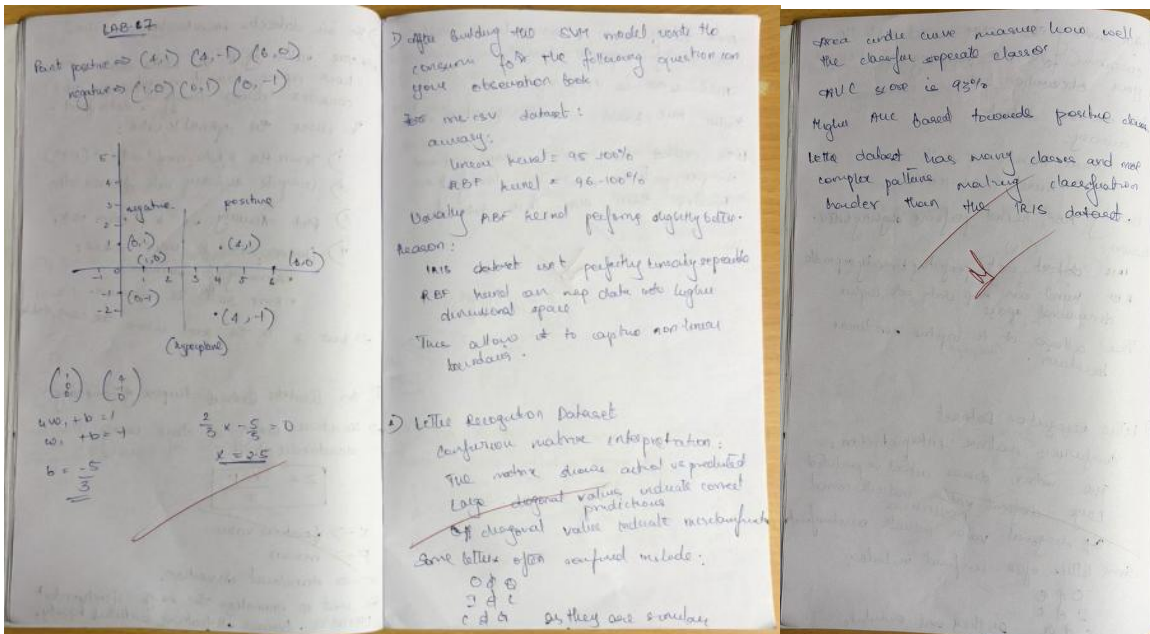
# Classification Report
print("Classification Report:\n", classification_report(y_test, y_pred))

```

Program 7

Build Support vector machine model for a given dataset

Screenshot



Code:

```
# SVM Classifier for IRIS Dataset (Linear & RBF Kernel)
# Import libraries
import pandas as pd
import numpy as np
import matplotlib.pyplot as plt

from sklearn.model_selection import train_test_split
from sklearn.svm import SVC
from sklearn.metrics import accuracy_score, confusion_matrix

# Load IRIS dataset
iris = pd.read_csv("iris.csv")

# Display dataset
print(iris.head())

# Features and target
X = iris.drop("species", axis=1)
y = iris["species"]

# Train-test split (80% training, 20% testing)
```

```

X_train, X_test, y_train, y_test = train_test_split(
    X, y, test_size=0.2, random_state=42
)

# -----
# SVM using RBF kernel
# -----
rbf_model = SVC(kernel='rbf')

rbf_model.fit(X_train, y_train)

y_pred_rbf = rbf_model.predict(X_test)

print("RBF Kernel Accuracy:", accuracy_score(y_test, y_pred_rbf))

print("Confusion Matrix (RBF):")
print(confusion_matrix(y_test, y_pred_rbf))

# -----
# SVM using Linear kernel
# -----
linear_model = SVC(kernel='linear')

linear_model.fit(X_train, y_train)

y_pred_linear = linear_model.predict(X_test)

print("Linear Kernel Accuracy:", accuracy_score(y_test, y_pred_linear))

print("Confusion Matrix (Linear):")
print(confusion_matrix(y_test, y_pred_linear))
# SVM Classifier for Letter Recognition Dataset + ROC + AUC
# Import libraries
import pandas as pd
import matplotlib.pyplot as plt

from sklearn.model_selection import train_test_split
from sklearn.svm import SVC
from sklearn.metrics import accuracy_score, confusion_matrix
from sklearn.preprocessing import label_binarize
from sklearn.metrics import roc_curve, auc

# Load dataset
letter = pd.read_csv("letter-recognition.csv")

print(letter.head())

```

```

# Features and target
X = letter.iloc[:,1:] # features
y = letter.iloc[:,0]  # letter label

# Train-test split (80% training, 20% testing)
X_train, X_test, y_train, y_test = train_test_split(
    X, y, test_size=0.2, random_state=42
)
# Train SVM model
model = SVC(kernel='linear', probability=True)
model.fit(X_train, y_train)
# Prediction
y_pred = model.predict(X_test)
# Accuracy
print("Accuracy:", accuracy_score(y_test, y_pred))
# Confusion Matrix
print("Confusion Matrix:")
print(confusion_matrix(y_test, y_pred))

# -----
# ROC Curve and AUC Score
# -----

# Convert labels to binary for ROC
classes = sorted(y.unique())
y_test_bin = label_binarize(y_test, classes=classes)

# Predict probabilities
y_score = model.predict_proba(X_test)

# ROC for first class
fpr, tpr, _ = roc_curve(y_test_bin[:,0], y_score[:,0])
roc_auc = auc(fpr, tpr)

# Plot ROC Curve
plt.figure()
plt.plot(fpr, tpr, label="ROC curve (AUC = %0.2f)" % roc_auc)
plt.plot([0,1], [0,1], linestyle='--')
plt.xlabel("False Positive Rate")
plt.ylabel("True Positive Rate")
plt.title("ROC Curve")
plt.legend()
plt.show()

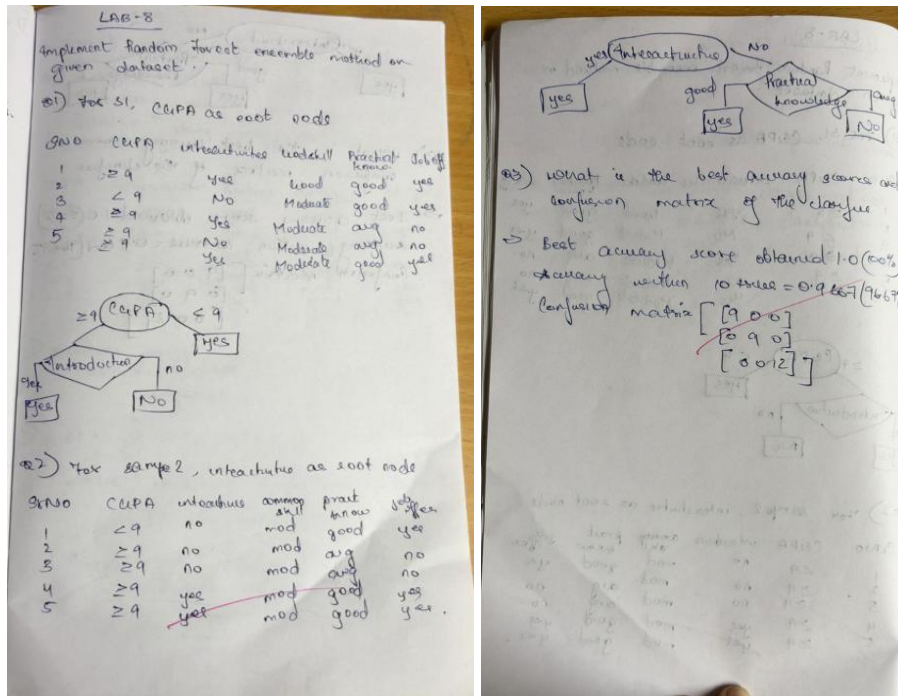
print("AUC Score:", roc_auc)

```

Program 8

Implement Random forest ensemble method on a given dataset.

Screenshot



Code:

```

# Import required libraries
import pandas as pd
from sklearn.model_selection import train_test_split
from sklearn.ensemble import RandomForestClassifier
from sklearn.metrics import accuracy_score

# Load dataset
data = pd.read_csv("iris.csv")

# Features and target
X = data.drop("species", axis=1) # assuming target column is named 'species'
y = data["species"]

# Train-test split
X_train, X_test, y_train, y_test = train_test_split(
    X, y, test_size=0.2, random_state=68
)

# -----
  
```



```

# 1. Default Random Forest (n_estimators = 10)
# -----
rf_default = RandomForestClassifier(n_estimators=10, random_state=42)
rf_default.fit(X_train, y_train)

y_pred_default = rf_default.predict(X_test)
default_score = accuracy_score(y_test, y_pred_default)

print("Accuracy with default (10 trees):", default_score)

# -----
# 2. Hyperparameter tuning (number of trees)
# -----
best_score = 0
best_n = 0

for n in range(1, 101): # try 1 to 100 trees
    rf = RandomForestClassifier(n_estimators=n, random_state=42)
    rf.fit(X_train, y_train)

    y_pred = rf.predict(X_test)
    score = accuracy_score(y_test, y_pred)

    if score > best_score:
        best_score = score
        best_n = n

print("Best accuracy:", best_score)
print("Best number of trees:", best_n)
from sklearn.metrics import confusion_matrix

# For default model
cm_default = confusion_matrix(y_test, y_pred_default)
print("Confusion Matrix (Default - 10 trees):\n", cm_default)

# For best model (inside loop OR after finding best_n)
rf_best = RandomForestClassifier(n_estimators=best_n, random_state=42)
rf_best.fit(X_train, y_train)

y_pred_best = rf_best.predict(X_test)
cm_best = confusion_matrix(y_test, y_pred_best)

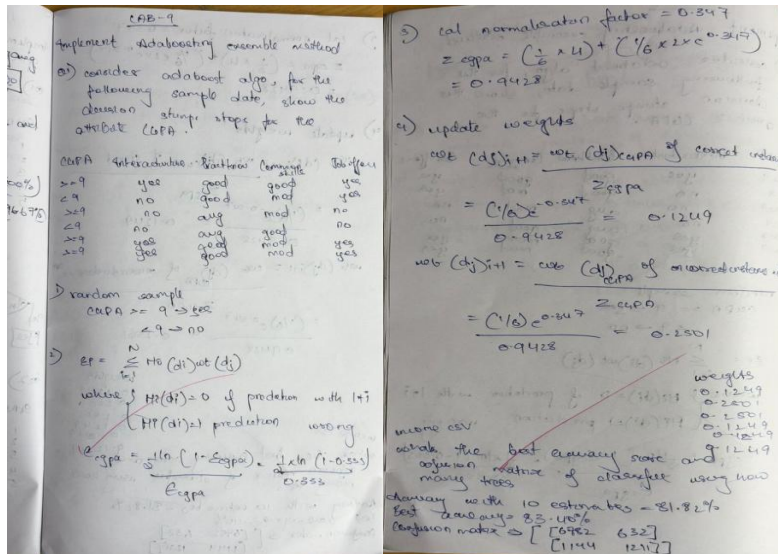
print("\nConfusion Matrix (Best Model):\n", cm_best)

```

Program 9

Implement Boosting ensemble method on a given dataset.

Screenshot



Code:

```
# Import libraries
import pandas as pd
from sklearn.model_selection import train_test_split
from sklearn.ensemble import AdaBoostClassifier
from sklearn.metrics import accuracy_score, confusion_matrix
```

```
# Load dataset
data = pd.read_csv("income.csv")
```

```
# Features and target
X = data.drop("income_level", axis=1)
y = data["income_level"]
```

```
# Train-test split
X_train, X_test, y_train, y_test = train_test_split(
    X, y, test_size=0.2, random_state=42
)
```

```
# -----
# 1. AdaBoost with 10 estimators
# -----
```

```

ada_default = AdaBoostClassifier(n_estimators=10, random_state=42)
ada_default.fit(X_train, y_train)

y_pred_default = ada_default.predict(X_test)
default_score = accuracy_score(y_test, y_pred_default)

print("Accuracy with 10 estimators:", default_score)

cm_default = confusion_matrix(y_test, y_pred_default)
print("Confusion Matrix (Default):\n", cm_default)

# -----
# 2. Hyperparameter tuning
# -----
best_score = 0
best_n = 0
best_cm = None

for n in range(1, 101):
    ada = AdaBoostClassifier(n_estimators=n, random_state=42)
    ada.fit(X_train, y_train)

    y_pred = ada.predict(X_test)
    score = accuracy_score(y_test, y_pred)

    if score > best_score:
        best_score = score
        best_n = n
        best_cm = confusion_matrix(y_test, y_pred)

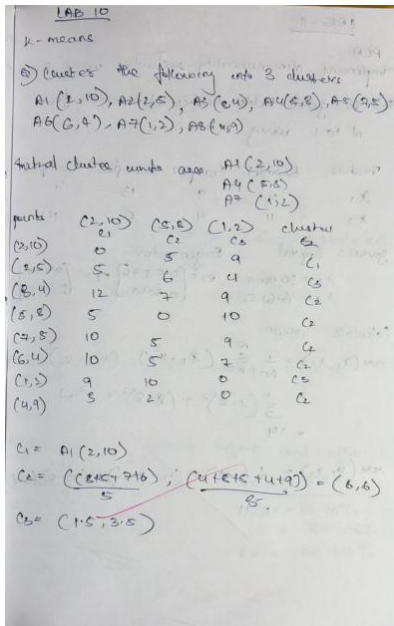
print("\nBest Accuracy:", best_score)
print("Best number of estimators:", best_n)
print("Confusion Matrix (Best Model):\n", best_cm)

```

Program 10

Build k-Means algorithm to cluster a set of data stored in a .CSV file.

Screenshot



Code:

```
# Import libraries
import pandas as pd
from sklearn.cluster import KMeans
from sklearn.preprocessing import StandardScaler
from sklearn.metrics import silhouette_score

# Load dataset
data = pd.read_csv("iris.csv")

# -----
# Preprocessing
# -----

# Drop target column if present (unsupervised learning)
if "species" in data.columns:
    X = data.drop("species", axis=1)
else:
    X = data

# Optional: Feature scaling (recommended for K-Means)
scaler = StandardScaler()
```

```

X_scaled = scaler.fit_transform(X)

# -----
# 1. Apply K-Means (k = 3)
# -----
kmeans = KMeans(n_clusters=3, random_state=42)
kmeans.fit(X_scaled)

labels = kmeans.labels_

print("Cluster labels:\n", labels)

# -----
# 2. Evaluate clustering
# -----
score = silhouette_score(X_scaled, labels)
print("Silhouette Score:", score)

# -----
# 3. Cluster centers
# -----
print("Cluster Centers:\n", kmeans.cluster_centers_)

```

Program 11

Implement Dimensionality reduction using Principal Component Analysis (PCA) method.

Screenshot

CAS-11

PCA
Implement dimensionality reduction using PCA

Q) In given table reduce dimension from 2 to 1 using PCA

Feature	Example 1	Example 2	Example 3	Example 4
X_1	4	8	13	7
X_2	11	4	5	14

given: Eig val
 $\lambda_1 = 50.5849$
 $\lambda_2 = 6.6151$

Eigen vector
 $e_1 = \begin{bmatrix} 0.5576 \\ 0.8303 \end{bmatrix}$ $e_2 = \begin{bmatrix} 0.8303 \\ -0.5576 \end{bmatrix}$

Calculate mean

$$\text{cov}(X_1, X_2) = \frac{1}{N-1} \sum_{k=1}^N (X_{1k} - \bar{X}_1)(X_{2k} - \bar{X}_2)$$

$$= \frac{1}{3} (4-8)^2 + (8-8)^2 + (13-8)^2 + (7-8)^2$$

$$= 14$$

$\text{cov}(X_1, X_2) = \frac{1}{3} (4-8) + (8-8.5) + (13-8)(5.5)$
 $= 11$

$$S = \begin{bmatrix} \text{cov}(X_1, X_1) & \text{cov}(X_1, X_2) \\ \text{cov}(X_2, X_1) & \text{cov}(X_2, X_2) \end{bmatrix} = \begin{bmatrix} 14 & 11 \\ 11 & 23 \end{bmatrix}$$

Eigen value
 $\lambda_1 = 50.5849$ $\lambda_2 = 6.6151$

Eigen Vector
 $e_1 = \begin{bmatrix} 0.5576 \\ 0.8303 \end{bmatrix}$

$$e_1^T = \begin{bmatrix} X_{1k} - \bar{X}_1 \\ X_{2k} - \bar{X}_2 \end{bmatrix} = \begin{bmatrix} 0.5576(4-8) - 0.8303(11-8.5) \\ 0.8303(4-8) + 0.5576(11-8.5) \end{bmatrix}$$

$$= \begin{bmatrix} -4.3062 \\ 5.7361 \end{bmatrix}$$

Feature	eX_1	eX_2	eX_3	eX_4
X_1	4	8	13	7
X_2	11	4	5	14
* principal component	-4.3062	5.7361	5.6929	-5.1238

Heatb cov

Report accuracy scores before and after

before PCA	after PCA
SVM = 86.4%	SVM = 85.86%
LR = 85.32%	LR = 85.526%
RF = 87.5%	RF = 85.89%
Best accuracy	

Sourav

Code:

```
# Import libraries
import pandas as pd
from sklearn.model_selection import train_test_split
from sklearn.preprocessing import OneHotEncoder, StandardScaler
from sklearn.compose import ColumnTransformer
from sklearn.pipeline import Pipeline
from sklearn.metrics import accuracy_score
from sklearn.decomposition import PCA
```

Models

```
from sklearn.svm import SVC
```

```

from sklearn.linear_model import LogisticRegression
from sklearn.ensemble import RandomForestClassifier

# Load dataset
data = pd.read_csv("heart.csv")

# -----
# Features and target
# -----
X = data.drop("HeartDisease", axis=1)
y = data["HeartDisease"]

# -----
# Column separation
# -----
categorical_cols = ["Sex", "ChestPainType", "RestingECG", "ExerciseAngina", "ST_Slope"]
numeric_cols = ["Age", "RestingBP", "Cholesterol", "FastingBS", "MaxHR", "Oldpeak"]

# -----
# Preprocessing
# -----
preprocessor = ColumnTransformer(
    transformers=[
        ("num", StandardScaler(), numeric_cols),
        ("cat", OneHotEncoder(), categorical_cols)
    ]
)

# -----
# Train-test split
# -----
X_train, X_test, y_train, y_test = train_test_split(
    X, y, test_size=0.2, random_state=42
)

# -----
# Models
# -----
models = {
    "SVM": SVC(),
    "Logistic Regression": LogisticRegression(max_iter=1000),
    "Random Forest": RandomForestClassifier()
}

# -----
# BEFORE PCA
# -----
print("---- BEFORE PCA ---- ")

```

```

best_score = 0

for name, model in models.items():
    pipe = Pipeline([
        ("preprocess", preprocessor),
        ("classifier", model)
    ])

    pipe.fit(X_train, y_train)
    y_pred = pipe.predict(X_test)
    score = accuracy_score(y_test, y_pred)

    print(f'{name}: {score}')

    if score > best_score:
        best_score = score

print("Best Accuracy before PCA:", best_score)

# -----
# AFTER PCA
# -----
print("\n---- AFTER PCA ----")

pca = PCA(n_components=0.95)

best_score_pca = 0

for name, model in models.items():
    pipe = Pipeline([
        ("preprocess", preprocessor),
        ("pca", pca),
        ("classifier", model)
    ])

    pipe.fit(X_train, y_train)
    y_pred = pipe.predict(X_test)
    score = accuracy_score(y_test, y_pred)

    print(f'{name}: {score}')

    if score > best_score_pca:
        best_score_pca = score

print("Best Accuracy after PCA:", best_score_pca)

```